

5. Program on demonstrations of Interface.

```
interface Inf1 {
    public int fact(int n);
}

interface Inf2 extends Inf1 {
    public float ncr(int n, int r);
}

public class Interfaceinherit implements Inf2 {
    public int fact(int n) {
        int i, f = 1;
        for (i = 1; i <= n; i++)
            f = f * i;
        return (f);
    }
    public float ncr(int n, int r) {
        float res;
        res = fact(n) / (fact(n - r) * fact(r));
        return res;
    }
    public static void main(String args[]) {
        Inf2 obj = new Interfaceinherit();
        // Interfaceinherit obj = new Interfaceinherit();
        System.out.println("factorial = " + (obj.fact(4)));
        System.out.println("ncr = " + (obj.ncr(4, 2)));
    }
}
```

Output:

```
C:\Users\reva00381\Desktop\Java\Java>javac Interfaceinherit.java
```

```
C:\Users\reva00381\Desktop\Java\Java>java Interfaceinherit
```

```
factorial =24
```

```
ncr =6.0
```

6. Program on Creation of Packages, importing the user defined Package.

//Create this java file and save as B_I.java

```
package pack;
public interface B_I
{
    public void getProd(int a, int b);
}
```

//Create this java file and save as Sum_A.java

```
package pack;
public class Sum_A implements B_I
{
    public void getProd(int a, int b)
    {
        System.out.println("Prod Of Two Numbers"+a+" and "+b+"="+a*b);
    }
    public void getSum(int a, int b)
```

```
    {  
    System.out.println("\nSum Of Two Numbers"+a+" and "+b+"="+a+b);  
    }  
}
```

//Create this java file and save as packdemo_1.java

```
import pack.Sum_A;  
class packdemo_1  
{  
    public static void main(String[] args)  
    {  
        Sum_A s=new Sum_A();  
        s.getSum(10,20);  
        s.getProd(2,6);  
    }  
}
```

Output:

C:\Users\reva00381\Desktop\Java>javac -d . B_I.java

C:\Users\reva00381\Desktop\Java>javac -d . Sum_A.java

C:\Users\reva00381\Desktop\Java>javac packdemo_1.java

C:\Users\reva00381\Desktop\Java>java packdemo_1

Sum Of Two Numbers10 and 20=30

Prod Of Two Numbers 2 and 6=12